Plant Disease Recognition

Section 1 Group 9
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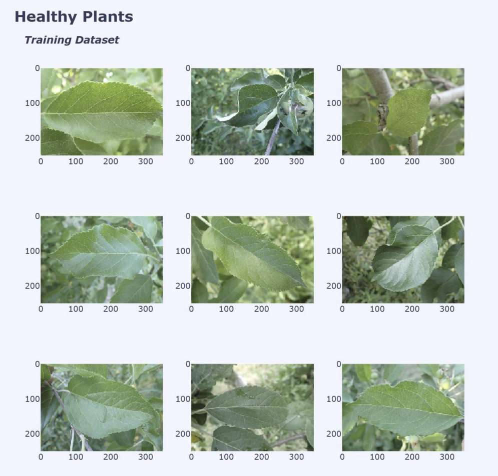
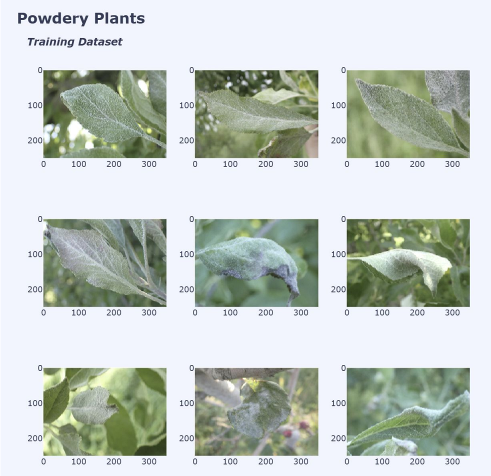
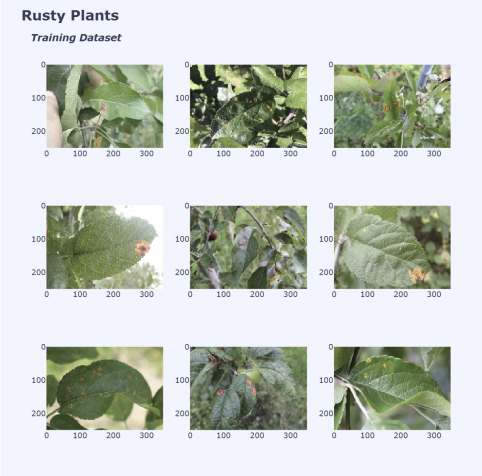
Table of Contents

Introduction

- Dataset = Plant Disease Recognition (Healthy, Powdery, Rusty Leaves)
- Feature Extraction
  - Edge Detection
  - Color Channels
  - Texture Detection
  - Blob Detection
  - Luminance Histogram
  - ResNet-50







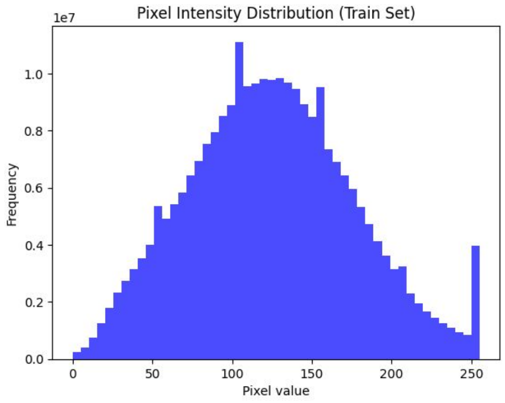
Dataset Explanation

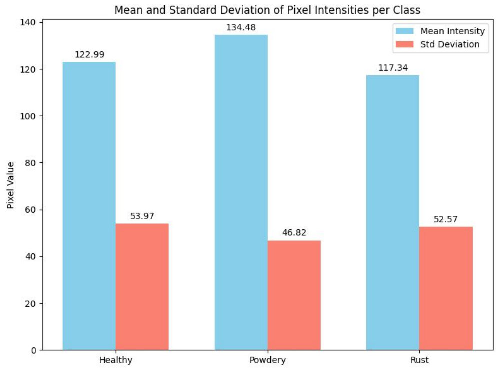
Dataset contains **1,530 pictures** of plant leaves divided into **3 classes**:

- **Healthy**: 528 total images
  - Train: 458
  - Validation: 20
  - Test: 50
- **Powdery**: 500 total images
  - Train: 430
  - Validation: 20
  - Test: 50
- **Rust**: 504 total images
  - Train: 434
  - Validation: 20
  - Test: 50

Image size varies between **~200 KB** and **1 MB**.

Resolution of most images is **4000x2672 pixels**.
Few outliers at **5184x3458** and **2592x1728 pixels**.





Data Preprocessing

- Removing background with **RemBG** preprocessing tool.
- Center-cropping images to a square **1:1 aspect ratio**.
- Normalizing images to a fixed resolution of **256×256 pixels**.

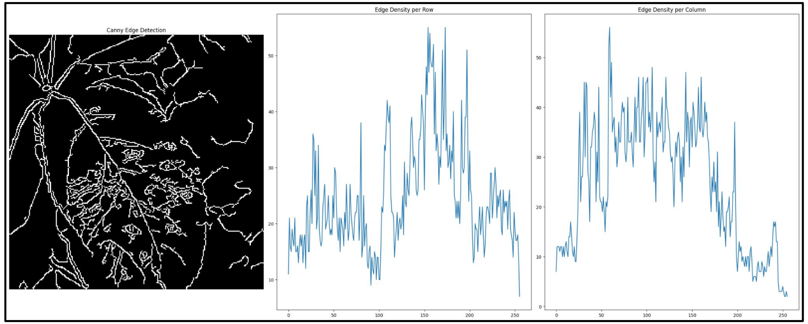
Feature #1: Edge Detection

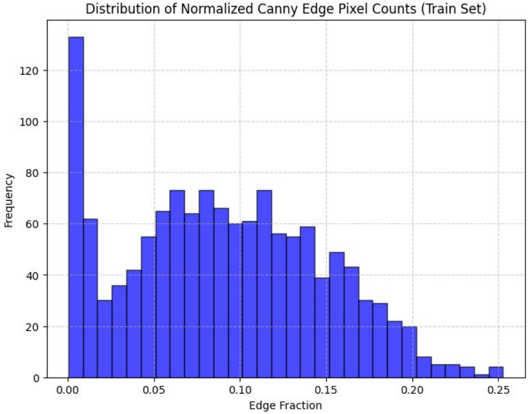
Why edge detection is important?

- **Powdery** surface is less defined with less sharp transitions between light and dark areas.
- **Rusty** leaves have sharper peaks and more rapid pixel intensity changes
- **Healthy** leaves have smoother edges and a more uniformly distributed pixel value

How to detect Edges?

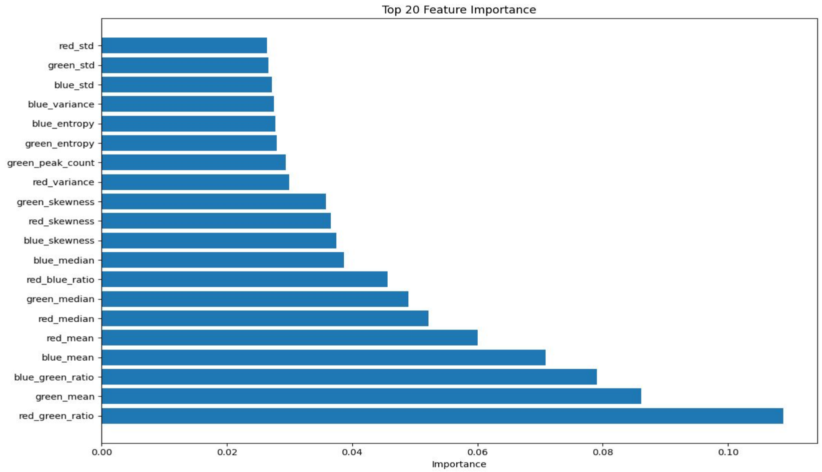
- Canny Edge Detection (CV2)
- Convert to Edge Pixel Count

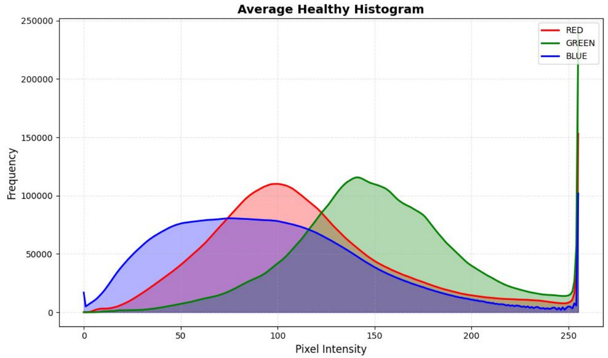


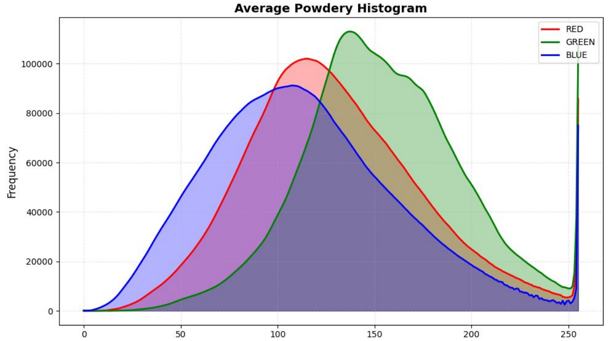


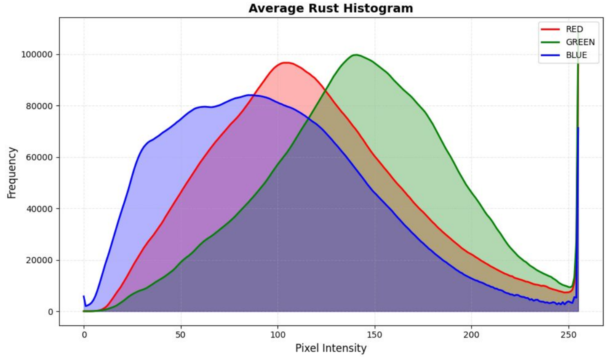
Feature #2: Color Channels

- Histogram of pixel intensity and Frequency generated for each color channel for each image
- Several Statistical Measures taken from RGB Color Channels
- Red-Green mean intensity Ratio, Mean of Green Channel among strongest predictors









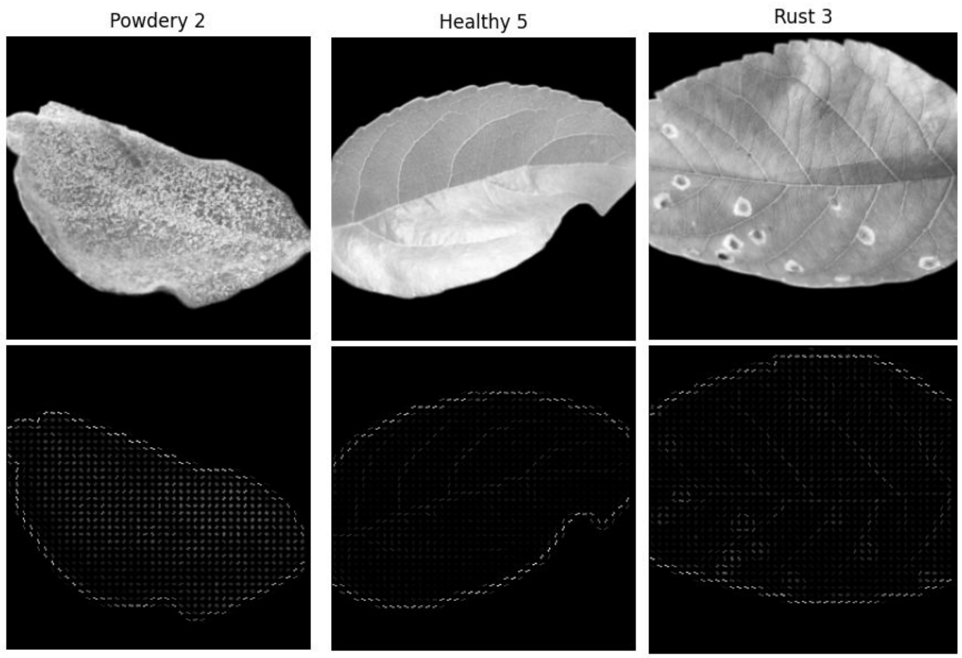
Feature #3: Texture Detection

Why texture is important?

- **Powdery** mildew has grainy patterns across leaf surface.
- **Rust** forms round pustules with distinct texture.
- **Healthy** leaves are smoother with uniform textures.

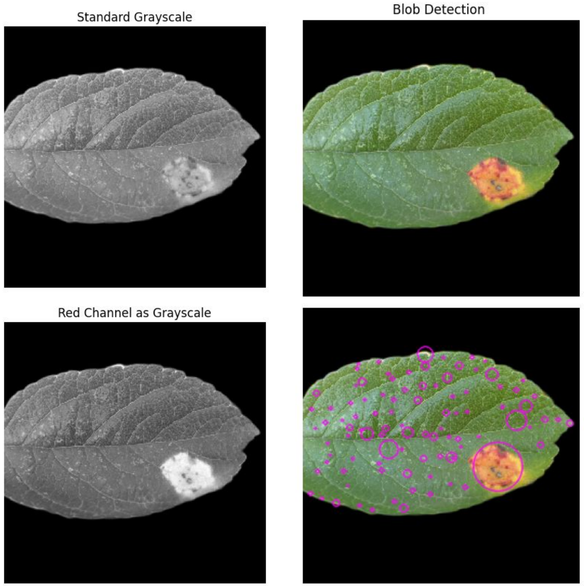
How to detect texture?

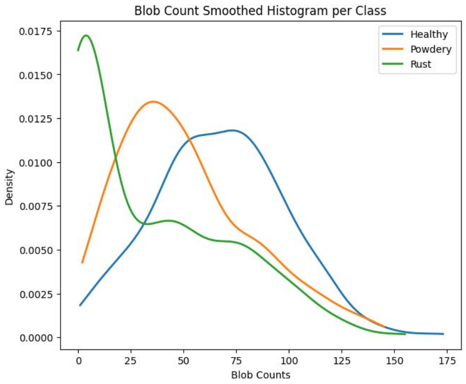
- Use **HOG** (Histogram of Oriented Gradients)

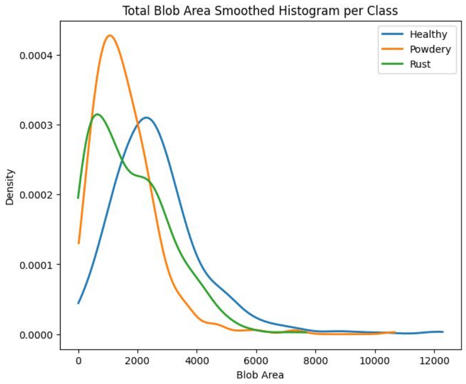


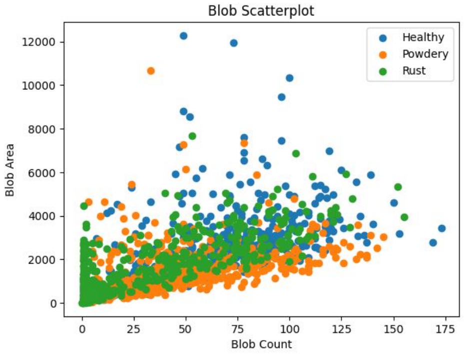
Feature #4: Blob Detection

- Targeting unhealthy spots on leaves
- Blob count and blob area from cv2 SimpleBlobDetector
- Blobs detected from red channel as white pixels



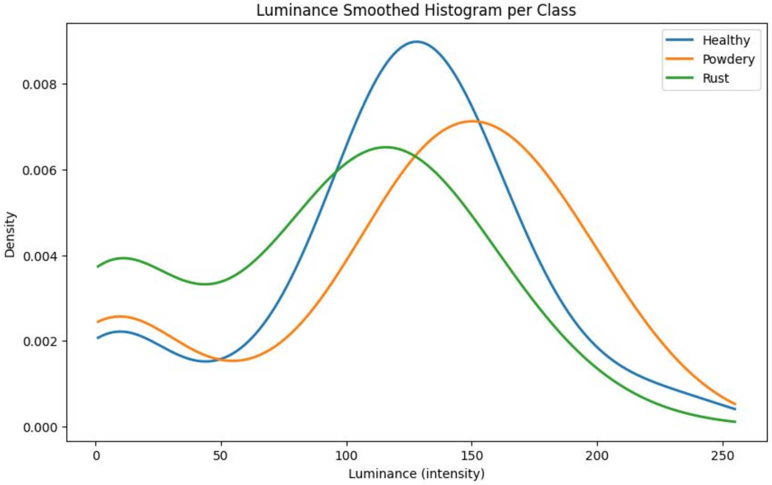


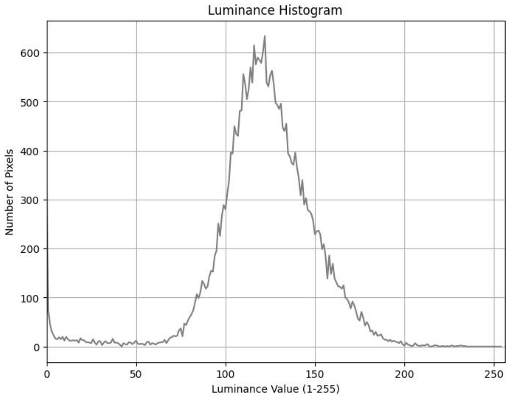




Feature #5: Luminance

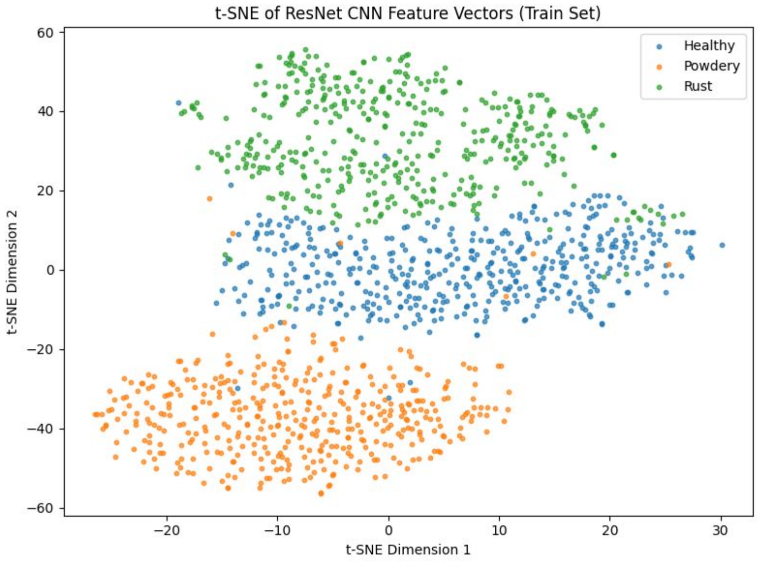
- Targeting brightness and reflectivity of leaves
- Histogram of grayscale pixel intensities 1-256
  - Excluding 0 intentionally due to background removal





Feature #6: ResNet50

- Complex, pretrained deep learning CNN
- Classification layer removed from top, last dense layer output used as feature vector
- Yielded best results when applied to images with backgrounds



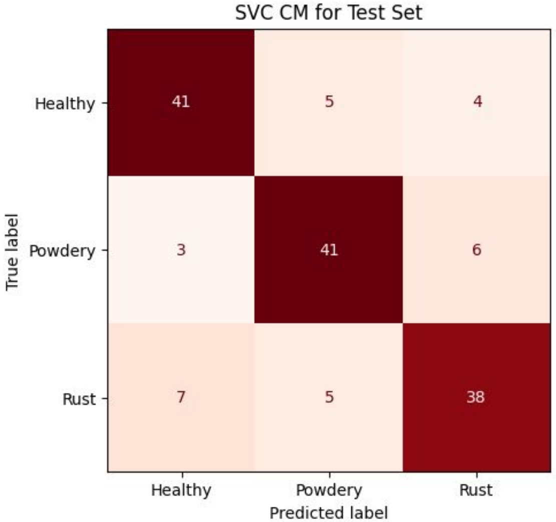
Model Selection

- Baseline
  - Grayscale images converted to a feature vector and fed into a logistic regression model
  - Classes were already balanced in raw dataset; better option than a majority class classifier
- Logistic Regression
  - Excels with meaningful features and is efficient to train and evaluate w/ validation set
- Support Vector Classifier
  - Works well when combining a multitude of features, especially HOG and PCA-reduced features.
- Random Forest
  - Accessible ensemble method (relatively low runtime) with built-in feature prioritization
  - Generally robust

Summary of Results

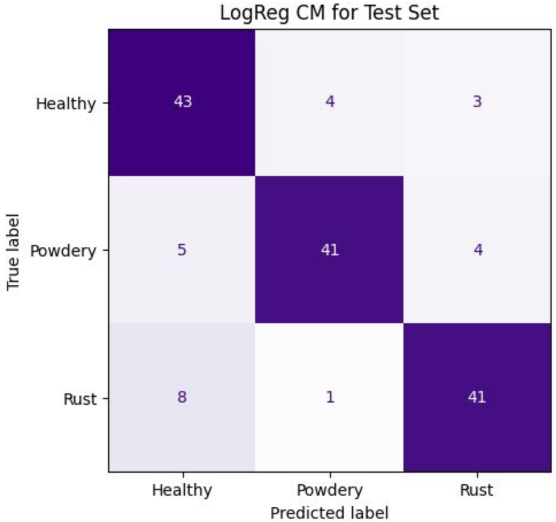
Model	Accuracy (Train/Val/Test)	AUC (Val/Test)	Runtime
Baseline LogReg	1.0 / 0.43 / 0.46	0.66 / 0.66	15.3 s
SVC w/o ResNet	0.76 / 0.78 / 0.76	0.94 / 0.90	1min 16s
SVC w/ ResNet	0.86 / 0.87 / 0.80	0.98 / 0.95	3min 22s
LogReg w/o ResNet	0.77 / 0.75 / 0.73	0.94 / 0.89	13.9 s
LogReg w/ ResNet	0.87 / 0.85 / 0.83	0.98 / 0.96	1min 9s
RF w/o ResNet	0.78 / 0.83 / 0.74	0.96 / 0.92	18.6 s
RF w/ ResNet	0.98 / 1.0 / 0.96	1.0 / 0.99	39.4 s

Final Model Results w/ ResNet (Test Set)



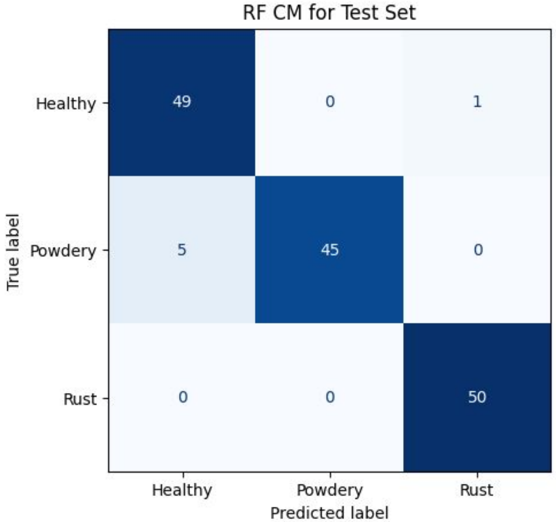
Test Results:

	precision	recall	f1-score	support
0	0.80	0.82	0.81	50
1	0.80	0.82	0.81	50
2	0.79	0.76	0.78	50
accuracy			0.80	150
macro avg	0.80	0.80	0.80	150
weighted avg	0.80	0.80	0.80	150



Test Results:

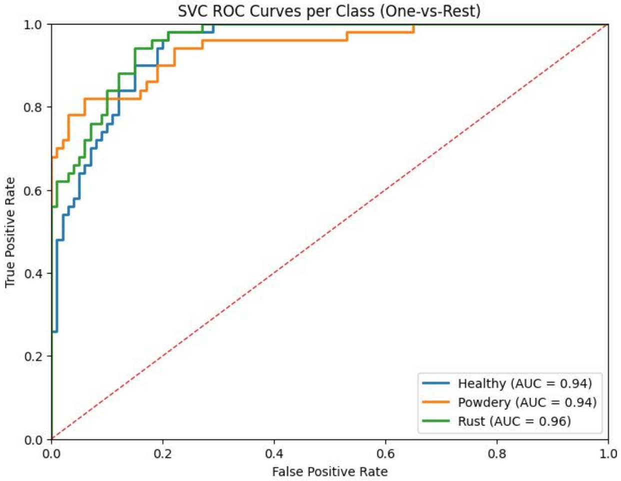
	precision	recall	f1-score	support
0	0.77	0.86	0.81	50
1	0.89	0.82	0.85	50
2	0.85	0.82	0.84	50
accuracy			0.83	150
macro avg	0.84	0.83	0.83	150
weighted avg	0.84	0.83	0.83	150

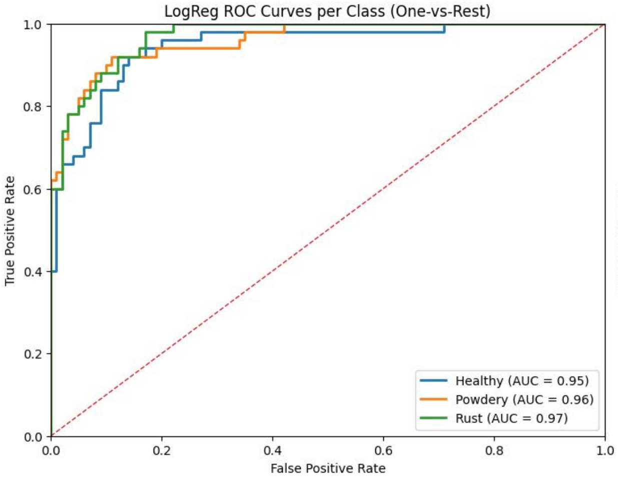


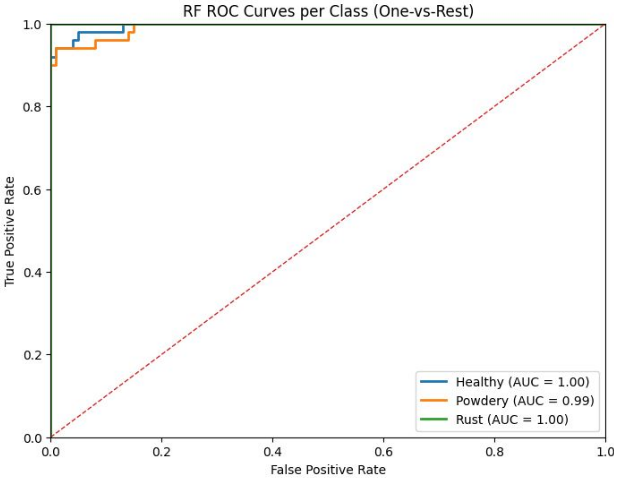
Test Results:

	precision	recall	f1-score	support
0	0.91	0.98	0.94	50
1	1.00	0.90	0.95	50
2	0.98	1.00	0.99	50
accuracy			0.96	150
macro avg	0.96	0.96	0.96	150
weighted avg	0.96	0.96	0.96	150

Final Model Results w/ ResNet (Test Set)







Potential Improvements

- Feature transformations (scaling, contrast enhancement, etc.)
- Feature hyperparameter tuning
- Test feature contribution; robust feature selection
- Improved background removal
  - Sometimes inconsistent due to variation in pictures
- More data; external datasets for testing the models
  - Repeat process for other plant species

Key Takeaways

- Decision tree ensemble model with deep learning feature responded very well to the problem
  - Decision trees excel at prioritizing some features over others; likely prioritized the deep learning feature
- Class balance, quality of data, background removal were important
  - Removal of noise and unnecessary information significantly improves performance
  - How pictures were taken affects how well features are detected (ex: poor focus on rusty/powdery surfaces can distort results)
- Not all features contribute equally to the model’s performance, but they take time to process which increases runtime
  - Analyzing feature contribution score is essential for building efficient and accurate model
  - Complex features take more processing time but may be less effective than simpler ones

Questions